

LAC MEMPHREMAGOG, QC, Canada

WMO: 710359

Lat: 45.2677N

Lon: 72.1617W

Elev: 209

StdP: 98.84

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.3	(c) -22.4	(d) -29.6	(e) 0.3	(f) -24.6	(g) -26.8	(h) 0.3	(i) -21.9	(j) 12.5	(k) -7.4	(l) 11.2	(m) -9.0	(n) 2.2	(o) 270	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	28.2	21.3	26.9	20.5	25.7	19.9	23.0	26.4	22.1	25.4	21.2	24.2	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.0	24.9	20.9	16.0	24.0	20.1	15.1	23.3	69.2	26.4	65.6	25.5	62.6	24.3	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.9	8.4	7.1		-28.2	30.8	3.8	1.0	-30.9	31.5	-33.1	32.1	-35.3	32.6	-38.1	33.3	
				-29.0	24.4	4.2	1.2	-32.0	25.2	-34.5	25.9	-36.8	26.6	-39.9	27.5	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-9.1	-7.9	-3.2	4.7	12.4	17.2	20.5	19.6	15.8	8.6	2.1	-4.4
	DBStd	11.52	7.46	6.56	6.28	4.65	4.14	3.47	2.76	2.77	3.96	4.62	5.02	6.13
	HDD10.0	2466	593	501	408	169	22	1	0	0	3	84	239	445
	HDD18.3	4530	852	734	666	409	189	62	10	19	97	303	486	703
	CDD10.0	1165	0	0	1	10	96	217	326	297	176	41	3	0
	CDD18.3	187	0	0	0	0	4	28	77	58	19	1	0	0
	CDH23.3	1008	0	0	0	2	34	152	434	289	95	1	0	0
	CDH26.7	131	0	0	0	0	3	19	65	32	11	0	0	0
(14) Wind	WSAvg	2.6	2.7	2.6	2.7	2.7	2.5	2.4	2.4	2.3	2.2	2.5	3.0	2.8
(15) Precipitation	PrecAvg	1182	89	67	78	86	97	114	131	117	101	110	93	99
	PrecMax	1415	156	94	114	189	158	169	226	189	215	233	143	203
	PrecMin	900	42	36	38	9	49	70	56	34	47	43	45	50
	PrecStd	130	29	18	19	43	27	31	40	39	42	52	25	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	8.8	14.8	21.7	26.7	28.8	30.1	29.2	28.3	22.7	17.4	10.2
		MCWB	6.5	5.9	9.4	14.2	17.4	21.4	23.3	22.6	21.5	17.0	13.8	8.3
	2%	DB	5.1	5.3	10.0	18.2	23.9	26.8	28.5	27.5	25.8	20.1	13.9	7.7
		MCWB	3.5	3.1	5.2	11.4	15.7	19.7	21.7	21.0	20.3	15.6	10.9	6.3
	5%	DB	3.3	3.4	7.3	15.0	21.8	25.1	27.0	26.2	24.1	17.9	11.6	5.5
		MCWB	2.1	1.6	4.0	9.0	15.0	18.7	21.0	20.2	19.2	14.2	8.9	4.4
	10%	DB	1.4	1.6	5.1	12.1	19.6	23.6	25.7	25.0	22.4	15.8	9.4	3.4
		MCWB	0.3	0.2	2.5	7.3	13.9	18.0	20.4	19.7	18.5	12.9	7.2	2.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.4	10.4	15.8	19.7	22.8	24.7	23.6	22.7	18.6	14.3	8.5
		MCDB	7.8	7.4	13.1	20.3	23.8	27.1	28.1	27.1	25.6	21.3	16.5	9.7
	2%	WB	4.0	3.5	6.3	12.7	18.0	21.6	23.5	22.4	21.5	16.6	11.5	6.5
		MCDB	4.7	4.9	8.7	17.1	21.6	24.9	26.6	25.8	24.6	18.7	13.8	7.4
	5%	WB	2.1	1.8	4.4	10.0	16.5	20.4	22.4	21.6	20.2	14.7	9.3	4.5
		MCDB	3.1	3.1	6.6	12.9	19.8	23.4	25.6	24.6	22.8	17.5	10.9	5.5
	10%	WB	0.3	0.4	2.8	8.0	14.9	19.2	21.5	20.7	19.0	13.1	7.4	2.4
		MCDB	1.4	1.5	4.9	11.2	18.1	22.1	24.5	23.6	21.8	15.5	9.2	3.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	10.3	9.5	9.9	8.8	8.4	8.5	9.1	7.9	7.4	6.9
		MCDBR	9.6	10.5	12.7	14.9	13.1	11.1	9.8	10.2	10.7	11.1	10.9	7.4
	5% WB	MCWBR	9.0	8.6	8.9	8.5	6.6	5.5	4.9	5.1	6.0	7.0	8.6	7.1
		MCWBR	9.2	9.8	10.5	13.0	10.3	9.1	8.3	8.3	9.4	9.6	9.8	7.3
(40) Clear-Sky Solar Irradiance	taub	taub	0.259	0.285	0.317	0.350	0.371	0.417	0.423	0.401	0.366	0.327	0.305	0.278
		taud	2.385	2.281	2.335	2.382	2.384	2.246	2.239	2.300	2.396	2.508	2.488	2.387
	Ebn at Noon	Ebn at Noon	867	897	916	910	898	853	842	848	852	845	803	797
		Edh at Noon	72	98	108	113	117	135	134	121	101	79	67	66
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.42	2.33	3.53	4.33	5.00	5.51	5.54	4.93	3.95	2.29	1.39	1.00
	RadStd	RadStd	0.18	0.26	0.27	0.50	0.52	0.43	0.43	0.26	0.32	0.22	0.18	0.09

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A

Nomenclature: See separate page